

=== GNB (1780330820) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
#
# srsRAN is free software: you can redistribute it and/or modify
# it under the terms of the GNU Affero General Public License as
# published by the Free Software Foundation, either version 3 of
# the License, or (at your option) any later version.
#
# srsRAN is distributed in the hope that it will be useful,
# but WITHOUT ANY WARRANTY; without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
# GNU Affero General Public License for more details.
#
# A copy of the GNU Affero General Public License can be found in
# the LICENSE file in the top-level directory of this distribution
# and at http://www.gnu.org/licenses/.
#

cu_cp:
  amf:
    addr: 10.53.1.1          # The address or hostname of the AMF.
    port: 38412
    bind_addr: 10.53.1.3     # A local IP that the gNB binds to for traffic from the AMF.
    supported_tracking_areas:
      - tac: 200
      plmn_list:
        - plmn: "00101"
      tai_slice_support_list:
        - sst: 1             # eMBB (enhanced Mobile Broadband) - 3GPP TS 23.501, Table 5.15.2.2-1
    inactivity_timer: 60     # Sets the UE/PDU Session/DRB inactivity timer to 60 seconds. Supported: [1 - 7200].

ru_sdr:
  device_driver: zmq         # The RF driver name.
  device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6 # Optionally pass argument
  srate: 11.52              # RF sample rate might need to be adjusted according to selected bandwidth.
  tx_gain: 75               # Transmit gain of the RF might need to adjusted to the given situation.
  rx_gain: 75               # Receive gain of the RF might need to adjusted to the given situation.

cell_cfg:
  dl_arfcn: 632628          # ARFCN of the downlink carrier (center frequency).
  band: 78                  # The NR band.
  channel_bandwidth_MHz: 40 # Bandwidth in MHz. Number of PRBs will be automatically derived.
  common_scs: 30            # Subcarrier spacing in kHz used for data.
  ssb_scs: 30              # Subcarrier spacing in kHz used for SSB.
  plmn: "00101"            # PLMN broadcasted by the gNB.
  tac: 200                  # Tracking area code (needs to match the core configuration).
  pdccch:
    common:
      ss0_index: 0          # Set search space zero index to match srsUE capabilities
      coreset0_index: 6     # Set search CORESET Zero index to match srsUE capabilities
    dedicated:
      ss2_type: common      # Search Space type, has to be set to common
      dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
  prach:
    prach_config_index: 1   # Sets PRACH config to match what is expected by srsUE
  pdsch:
    mcs_table: qam64        # Sets PDSCH MCS to 64 QAM
  pusch:
    mcs_table: qam64        # Sets PUSCH MCS to 64 QAM

log:
  filename: /tmp/gnb.log    # Path of the log file.
```

```
all_level: info                # Logging level applied to all layers.
hex_max_size: 0

pcap:
  mac_enable: false            # Set to true to enable MAC-layer PCAPs.
  mac_filename: /tmp/gnb_mac.pcap # Path where the MAC PCAP is stored.
  ngap_enable: true            # Set to true to enable NGAP PCAPs.
  ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.
  e2ap_enable: true            # Set to true to enable E2AP PCAPs.
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap # Path where the DU E2AP PCAP is stored.
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.
  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.

metrics:
  layers:
    enable_rlc: true           # Enable RLC metrics
    enable_sched: true         # Enable DU scheduler metrics
  periodicity:
    du_report_period: 500      # Set DU statistics report period to 500ms
```

=== UE (1780330820) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 11.52e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=11.52e6

[rat.eutra]
nof_carriers = 0

[rat.nr]
bands = 78
nof_carriers = 1
dl_nr_arfcn = 632628

max_nof_prb = 217
nof_prb = 217

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 112233445566778899aabbccddeeff00
k = 00aabbccddeeff112233445566778899
imsi = 0010100000000001
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER (1780330820) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
#
# srsRAN is free software: you can redistribute it and/or modify
# it under the terms of the GNU Affero General Public License as
# published by the Free Software Foundation, either version 3 of
# the License, or (at your option) any later version.
#
# srsRAN is distributed in the hope that it will be useful,
# but WITHOUT ANY WARRANTY; without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
# GNU Affero General Public License for more details.
#
# A copy of the GNU Affero General Public License can be found in
# the LICENSE file in the top-level directory of this distribution
# and at http://www.gnu.org/licenses/.
#
```

services:

```
5gc:
  container_name: open5gs_5gc
  build:
    context: open5gs
    target: open5gs
    args:
      OS_VERSION: "22.04"
      OPEN5GS_VERSION: "v2.7.0"
  env_file:
    - ${OPEN_5GS_ENV_FILE:-open5gs/open5gs.env}
  privileged: true
  ports:
    - "9898:9999/tcp"
  # Uncomment port to use the 5gc from outside the docker network
  # - "38412:38412/sctp"
  # - "2152:2152/udp"
  command: 5gc -c open5gs-5gc.yml
  healthcheck:
    test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
    interval: 3s
    timeout: 1s
    retries: 60
  networks:
    ran:
      ipv4_address: ${OPEN5GS_IP:-10.53.1.1}
```

```
gnb:
  container_name: srsran_gnb
  # Build info
  image: srsran/gnb
  build:
    context: ..
    dockerfile: docker/Dockerfile
    args:
      OS_VERSION: "24.04"
  # privileged mode is required only for accessing usb devices
  privileged: true
  # Extra capabilities always required
  cap_add:
    - SYS_NICE
    - CAP_SYS_PTRACE
  volumes:
    # Access USB to use some SDRs
```

```

- /dev/bus/usb/:/dev/bus/usb/
# Sharing images between the host and the pod.
# It's also possible to download the images inside the pod
- /usr/share/uhd/images:/usr/share/uhd/images
# Save logs and more into gnb-storage
- gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
  - gnb_config.yml
  - gnb_compose_config.yml
# Customize your desired network mode.
# current netowrk configuration creastes a private netwoek with both containers attached
# An alterantive would be `network: host`. That would expose your host network into the container. It's the easie
networks:
  ran:
    ipv4_address: ${GNB_IP:-10.53.1.3}
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

configs:
  gnb_config.yml:
    file: ${GNB_CONFIG_PATH:-./gnb.yaml} # Path to your desired config file
  gnb_compose_config.yml:
    content: |
      cu_cp:
        amf:
          addr: ${OPEN5GS_IP:-10.53.1.1}
          bind_addr: 0.0.0.0
        metrics:
          autostart_stdout_metrics: true
          enable_json: true
        remote_control:
          bind_addr: 0.0.0.0
          enabled: true

volumes:
  gnb-storage:

networks:
  ran:
    ipam:
      driver: default
      config:
        - subnet: 10.53.1.0/24
  metrics:
    ipam:
      driver: default
      config:
        - subnet: 172.19.1.0/24

```

=== NOTAS (1780330820) ===

1. ****PLMN Coherence (Rule 1):**** MCC "001", MNC "01" applied to gNB (plmn) and UE (IMSI). TAC 200 applied to gNB.
2. ****Frequency Coherence (Rule 2):**** dl_arfcn 632628 is within Band 78 range [620000, 653333].
3. ****SCS Coherence (Rule 3):**** common_scs and ssb_scs are both set to 30 KHz.
4. ****ZMQ Ports (Rule 4):**** Ports 2000 and 2001 are crossed between gNB and UE device_args.
5. ****bind_addr (Rule 5):**** gNB standalone config uses 10.53.1.3, docker-compose override uses 0.0.0.0.
6. ****ru_sdr (Rule 6):**** The ru_sdr block in gNB is copied exactly from the CAG template. This means srates is 11.52e6.
7. ****Bandwidth vs. Sample Rate:**** The requested 40 MHz bandwidth with 30 KHz SCS typically requires a sample rate of 11.52 MHz.
8. ****SST for eMBB:**** SST 1 for eMBB was applied to gNB, with a comment referencing 3GPP TS 23.501, Table 5.15.2.2-1.
9. ****UE PRB:**** max_nof_prb and nof_prb for UE are set to 217, corresponding to 40 MHz bandwidth with 30 KHz SCS.
10. ****docker-compose.yml:**** OPEN5GS_IP and GNB_IP environment variables are set to 10.53.1.1 and 10.53.1.3 respectively.